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1 Chapter 08 — Safe Exploitation in Imperfect-Information Games

Games: Kuhn Poker (3-card, 2-player) and Leduc Hold'em (6-card, 2-round) — the smallest exact-solvable imperfect-information testbeds, reusing the Chapter 07 stack wholesale (the Chapter 02 Kuhn engine + CFR solver, the Chapter 03 Leduc engine, and Chapter 07's exact best-response / exploitability code and opponent-type zoo).

PhD connection: the **second half of Contribution #1 (Behavioral Adaptation Framework)** and the launch point for **Contribution #2 (Multi-Agent Safe Exploitation)**. Chapter 07 built the *sensor* (an opponent model that turns observed actions into a strategy estimate); Chapter 08 builds the *actuator* — turning that estimate into profit **without becoming exploitable** — and pins down exactly where the two-player zero-sum assumption enters the safety guarantees (the thesis attack point).

Scope of results: every number in this report is **measured from a real run** and read from the run artifacts under `implementation/step08/implementation/results/`, `.../plots/`, and `.../exploration/figures/`. Wherever possible a simulated number is bracketed by an *exact* analytical reference (the Nash game value, the exact best-response value) rather than another simulation. Two Phase-4 predictions were **contradicted** by the runs; per the project workflow they are kept as stated and reconciled with what actually happened (§7, §10).

How to read this report. Both parts follow the same arc: **what we test** → **how we test it** → **results** → **conclusions**. **Part I (§1–§4)** makes the exploitation-safety tradeoff visible on Kuhn with two small seeded sweeps. **Part II (§5–§13)** evaluates the full system — one sequence-form LP engine and five safe-exploitation solver families — on Kuhn and Leduc against exact yardsticks. §14 lists reproduction commands.

2 PART I — THE TRADEOFF, MADE VISIBLE ON KUHN

2.1 1. What this chapter is about

A **Nash-equilibrium** strategy never loses in the long run against *anyone*, but it also never *punishes* a weak opponent — it plays identically against a world champion and against someone who folds every time you bet. Chapter 07 showed that a good opponent model unlocks 0.1–0.3 per hand against exploitable opponents, but also that best-responding to an imperfect model can open a leak in your own play (the continuous model *lost* to Nash on Leduc). **Safe exploitation** is the discipline that resolves that tension: lean toward exploiting the model, but bound how much a worst-case adversary can punish you.

Formally, every method in this chapter is the **same constrained optimization**:

maximize the hero’s expected value against the opponent model, **subject to** a safety floor on the hero’s worst-case value.

The expected value is *linear* in the hero’s sequence-form realization plan, so this is a linear program; the five method families differ only in **what the safety floor is**:

Method	Safety floor	Origin
Restricted Nash Response (RNR)	tunable via a parameter p (Nash at $p=0$ → best response at $p=1$)	Johanson 2007
Best equilibrium (ganzfried)	$\geq$ the Nash game value v^* , imposed on every hand’s strategy	Ganzfried & Sandholm 2015 (their best-equilibrium baseline)
Prime-safe	$\geq v^* - \varepsilon$, with ε = the baseline’s own exploitability	Jeary & Turrini 2023
SES (subgame)	$\geq$ the blueprint value, enforced <i>locally</i> on one subgame via a gadget (implemented here as Ge et al.’s adaptation safety)	Liu et al. 2022

Method	Safety floor	Origin
Adaptation safety	worst-case $\geq$ blueprint worst-case (i.e. no more exploitable than the blueprint)	Ge et al. 2024

A note on “best equilibrium”. Ganzfried & Sandholm define safety over the repeated game (at least v^* per hand in expectation). Imposed on each hand’s strategy, as here, the floor admits only equilibrium strategies, so the LP returns the equilibrium that does best against the model: their *best-equilibrium* baseline, not their safe-exploitation algorithms. Those (RWYWE, BEFFE) deviate beyond equilibrium by risking only the gifts already banked: RWYWE is the same LP with a per-hand floor $v^* - k_t$ that moves with the banked gifts. They are provably safe and can exploit more. Chapter 15 implements RWYWE on this LP and runs it under Chapter 14’s protocol: it gains +0.062 (Kuhn) and +0.113 (Leduc) chips per hand over the blueprint, against +0.020 and +0.047 for best equilibrium, and none of its 120 matches under teaching attacks ends below the game value. That is only 14–30 % of the gain of RNR at $p = 0.5$, because few gifts are provable when the opponent’s cards are seen only at showdown.

Part I probes the *naive* end of this menu — a behavioral blend of Nash and best response — precisely to show why the principled LP methods of Part II are needed. All Part I numbers are measured from seeded Kuhn runs; the hero is player 0, whose Kuhn game value is $-1/18 \approx -0.056$ (so “exploitation” means beating that value against a given opponent, and absolute numbers can stay negative against a tight opponent).

2.2 2. Experiment 1 — the naive Nash/best-response frontier

What we test. If you simply blend Nash and the full best response to a weak opponent — $(1-\lambda) \cdot \text{Nash} + \lambda \cdot \text{BR}$ — what tradeoff curve do you trace between *profit* and *your own exploitability*?

How. Sweep λ from 0 (pure Nash) to 1 (pure best response) against the very exploitable `TightPassive` type, and compute, **exactly**, the hero’s profit (EV vs `TightPassive`) and worst-case loss (game value minus the hero’s worst-case value). Data: `exploration/figures/pareto_curve_kuhn.json`.

λ (weight on BR)	profit (EV vs <code>TightPassive</code>)	worst-case loss (exploitability)
0.0 (Nash)	−0.047	0.001
0.3	+0.017	0.133
0.5	+0.060	0.222
0.7	+0.103	0.311

λ (weight on BR)	profit (EV vs TightPassive)	worst-case loss (exploitability)
1.0 (full BR)	+0.167	0.444

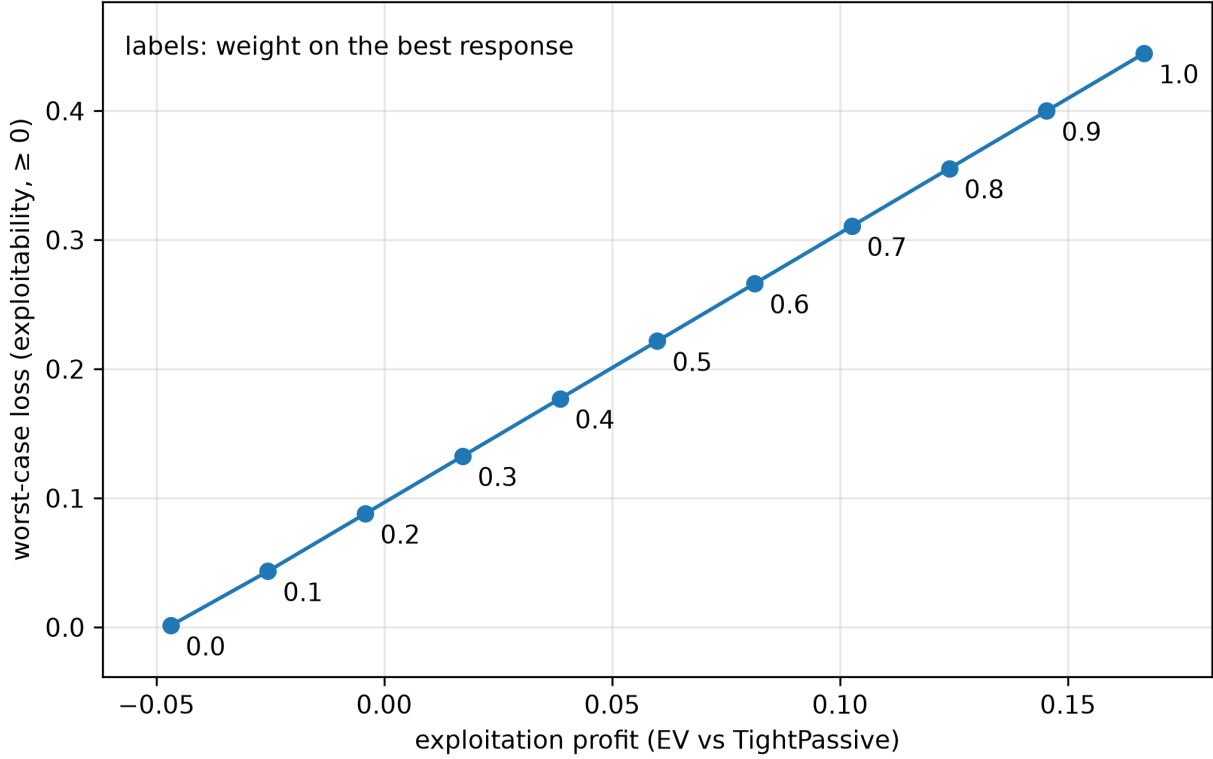


Figure 1: The naive Nash/best-response blend traces a smooth exploitation-safety frontier on Kuhn: every step toward more profit buys a near-proportional increase in your own worst-case loss.

Results. The blend traces a smooth, monotone line: each increment of profit costs a near-proportional increment of exploitability. Full best response wins the most (+0.167) but is maximally exploitable (worst-case loss 0.444 — i.e. an adversary can drive the hero down to -0.5); Nash is unexploitable but wins nothing extra.

Conclusion. The tradeoff is real and continuous, but a uniform blend is a *blunt* instrument — it scales deviation everywhere at once. The value of Part II’s LP methods is that they choose *where* to deviate, buying the same profit at less exploitability (§7). This curve is the baseline they must beat.

2.3 3. Experiment 2 — the naive RNR budget sweep (and a flag)

What we test. The same blend, re-parameterized as the raw step’s informal “Restricted Nash Response”: scale deviation by p and watch profit and exploitability grow. Is there a *budget* of cheap early exploitation?

How. Sweep $p \in \{0, \dots, 1\}$ against TightPassive; record profit and exploitability. Data: `exploration/figures/rnr_playground_kuhn.json`.

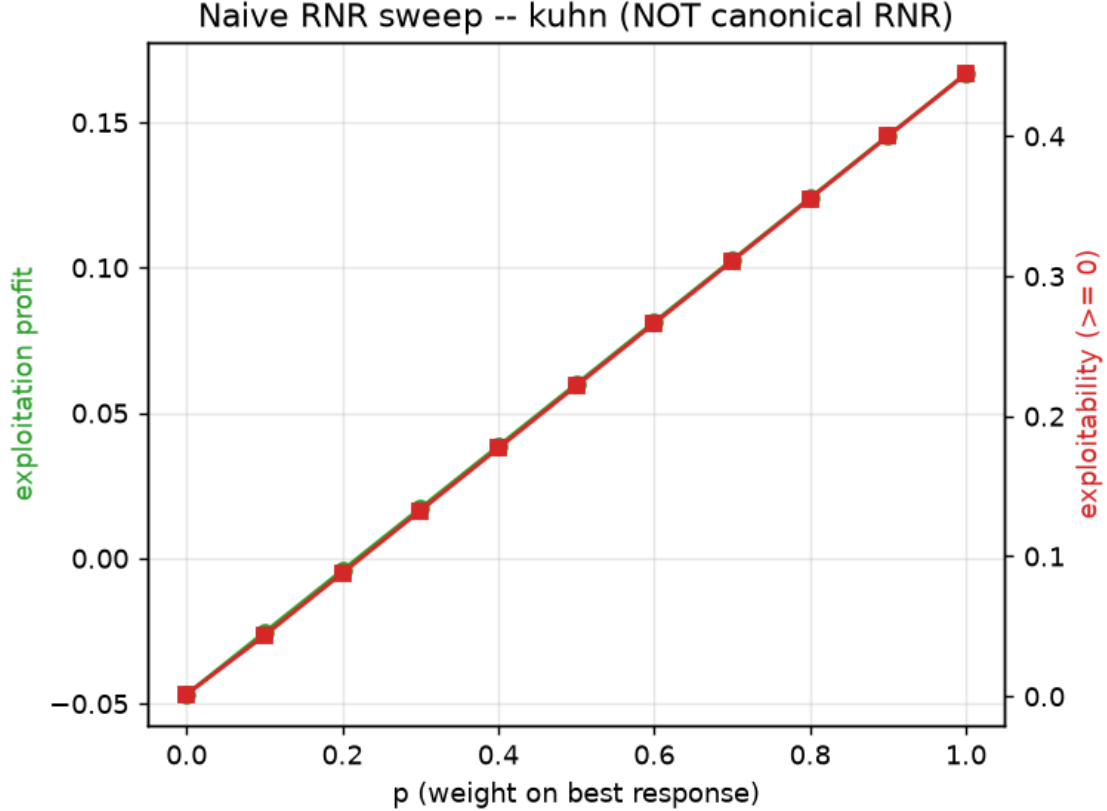


Figure 2: Naive RNR sweep on Kuhn: profit (left axis) and exploitability (right axis) both rise with p . This is the raw step’s informal blend — NOT Johanson’s canonical RNR, which Part II shows behaves very differently.

Results. Profit rises smoothly from -0.047 ($p=0$) to $+0.167$ ($p=1$); exploitability rises in lockstep from ≈ 0 to 0.444 . There is a modest early “budget”: the first increments of p buy profit while exploitability is still small.

Flag (naive $\neq$ canonical RNR). This p -blend is the raw step’s Day-2 *description*, and it is a fine intuition tool, but it is **not** Johanson’s Restricted Nash Response. Canonical RNR solves for the *equilibrium against a p -restricted opponent* — a max-min linear

program, implemented in Part II. Part II (§7) shows the two behave **completely differently** on Kuhn: the canonical version is bang-bang, not smooth. We keep both and label them, rather than conflate them.

Conclusion. The “budget” intuition holds for the naive blend, but Part II shows it is not how the principled algorithm behaves in a small game — a concrete example of why an intuition sweep must not be mistaken for the algorithm it illustrates.

2.4 4. What Part I establishes

1. **The exploitation-safety tradeoff is a real, measurable curve** (Exp. 1): more profit $\Rightarrow$ more of your own exploitability, near-proportionally, for the naive blend.
 2. **Full best response is dangerous** — maximally profitable against the model but maximally exploitable (worst-case -0.5 on Kuhn), which is the entire reason a safety floor is needed.
 3. **An intuition sweep is not the algorithm** (Exp. 2): the naive p-blend is smooth, but the canonical RNR it stands in for is not (§7). Part II replaces the blunt blend with LP methods that spend the exploitability budget *where it buys the most profit*.
-

3 PART II — FULL SYSTEM EVALUATION ON KUHN & LEDUC

3.1 5. What we test, and how

Part II evaluates the complete safe-exploitation system — **one sequence-form LP engine** and **five solver families** built on it — on both games, against exact analytical yardsticks.

The one engine. A HeroTreeplex enumerates the hero’s sequence-form realization plan (reusing Chapter 07’s treeplex), turns “EV against a fixed opponent” into a linear objective $c \cdot x$, and imposes the safety floor by **constraint generation (a double-oracle / cutting-plane loop)**: solve the LP $\rightarrow$ read the hero policy $\rightarrow$ call Chapter 07’s **exact best response** as the worst-case opponent $\rightarrow$ if the worst case is below the floor, add the linear cut and re-solve. The five families differ only in the floor (§1). This means one validated primitive (Chapter 07’s exact best response) powers both the objective and every safety check.

Exact yardsticks. Both games are small enough to solve exactly, so every solver’s reported profit and worst-case are computed on the full tree — no sampling. The two anchors are the **Nash game value** (v^* — the safe baseline) and the **exact full best-response value** (the exploitation ceiling). A safe method sits between them; an unsafe one has a worst-case below v^* .

The methods (string ids as they appear in the result tables): **nash** (baseline, no adaptation), **full_br** (maximize EV vs the model, no safety), **rnr_0.5** (canonical RNR at $p=0.5$), **ganzfried** (best equilibrium: $\text{floor} = v^*$), **prime_safe** ($\text{floor} = v^* - \epsilon$), **adaptation** ($\text{floor} = \text{blueprint worst-case}$), **ses_subgame** (subgame gadget; on Kuhn the subgame is the whole game, so it coincides with an adaptation solve).

The three studies.

1. **Exact method $\times$ opponent table** — for each opponent type, solve each method against a *perfect* model of that type and report exploitation EV (profit), worst-case value (safety), and whether the worst-case meets the Nash floor.
2. **The efficient frontier** — canonical RNR swept over p , the naive blend, and the best-equilibrium / prime-safe / adaptation operating points, all on one exploitation-vs-exploitability plot.
3. **Teaching attack (online)** — a deceptive opponent baits with a weak style then reveals a strong one; a Chapter 7 model feeds each solver every k hands; we track realized profit and safety-violation counts.

Configurations and runtimes (measured). Kuhn **smoke** (30 000 CFR iters): **1.4 s**. Kuhn **scale** (200 000 CFR iters, adds **AlwaysBet/AlwaysPass/ses_subgame**, teaching attack over 5 seeds $\times$ 20 000 hands): **10.3 s**. Leduc **bounded_scale** (a human-added config: a 40-iteration cap on constraint generation, `leduc_tol` = 0.01, SES on the King-flop subgame): **minutes** (individual SES cells 10–79 s). Everything is CPU / LP-bound — CFR, full-tree best response, and small SciPy HiGHS linear programs — so the GPU is irrelevant, as predicted. Measured game values: Kuhn $v^* = -0.0556$ ($\approx -1/18$ ✓), Leduc $v^* = -0.0862$.

Not captured. No `validate.py` PASS/FAIL log and no OpenSpiel cross-check artifact were saved from these runs, so those specific harness outputs are not reported; the exact tables below substantiate most validation targets directly (§11).

3.2 6. The exact method $\times$ opponent table (Kuhn)

The core deliverable: each method solved against a perfect model of each opponent, scored on profit (EV) and safety (worst-case), with the Nash floor at $v^* = -0.056$. From `results/kuhn_scale.json`.

Opponentmetric	nash	full_br	rnr_0.5	best eq.	prime_safe	adaptation	ses_subgame
TightPassive	−0.047	+0.167	−0.044	−0.044	−0.040	−0.040	−0.040
worst-case	−0.056	−0.500	−0.056	−0.056	−0.063	−0.063	−0.063

Opponent	metric	nash	full_br	rn timer_0.5	best eq.	prime_safe	adaptation	ses_subgame
Loose Agg.	EV	+0.118	+0.333	+0.278	+0.131	+0.151	+0.151	+0.151
	worst-case	−0.056	−0.167	−0.111	−0.056	−0.063	−0.063	−0.063
Always BeV	EV	+0.113	+0.333	+0.333	+0.115	+0.131	+0.131	+0.131
	worst-case	−0.056	−0.167	−0.167	−0.056	−0.063	−0.063	−0.063
Always Pass (most ex-ploitable)	EV	+0.146	+0.975	+0.975	+0.222	+0.266	+0.266	+0.266
	worst-case	−0.056	−0.500	−0.333	−0.056	−0.063	−0.063	−0.063
Nash (control)	EV	−0.056	−0.055	−0.056	−0.056	−0.055	−0.055	−0.055
	worst-case	−0.056	−0.333	−0.056	−0.056	−0.063	−0.063	−0.063

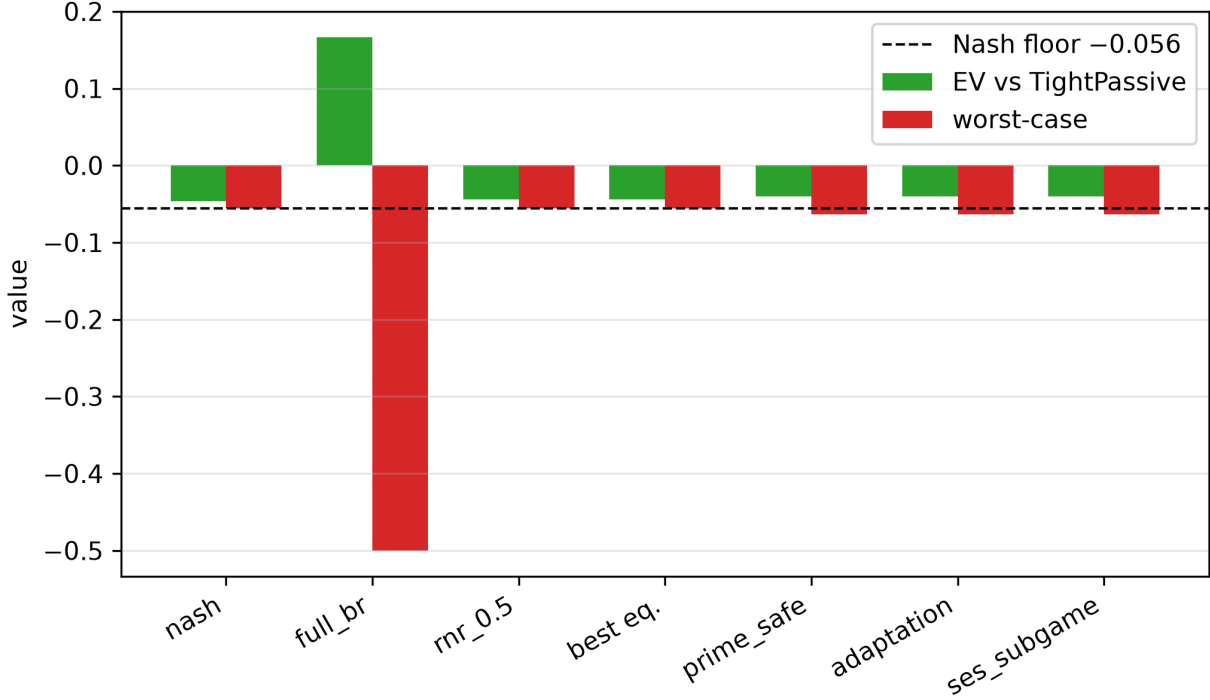


Figure 3: Methods vs TightPassive (Kuhn): green = EV vs the opponent, red = worst-case value, dashed = the Nash floor. Only full_br’s worst-case (red) plunges far below the floor; every other method hugs it.

Results — three carry the message.

1. **full_br is the cautionary tale.** It always wins the most against a fixed model (up to **+0.975** vs AlwaysPass), but its worst-case collapses to **−0.5** — an adversary can punish it catastrophically. It is the only method unsafe against *every* opponent (rnr_0.5 is unsafe against three). This is the danger the whole chapter exists to prevent.
2. **Best equilibrium (ganzfried) is safe and still profitable — but only slightly.** Against *every* opponent its worst-case stays at the Nash floor (safe within 1e-3) **and** it beats Nash’s own EV on every exploitable type — most strikingly **+0.222 vs Nash’s +0.146** against AlwaysPass, and +0.131 vs +0.118 against LooseAggressive. You can exploit while provably never dropping below equilibrium value, but only a little: against the four exploitable types in the table the gain over Nash is 0.002–0.076 per hand, 1–9 % of what the full best response gains, because a per-hand floor at v^* admits only equilibrium strategies (§1).
3. **rnr_0.5 is safe only for mild leaks.** Against TightPassive it is safe, but against the highly exploitable AlwaysPass/AlwaysBet it has *already jumped* to full best response (EV identical to full_br, worst-case −0.33 / −0.17, unsafe). The RNR transition is **opponent-dependent** — a single global p is not a safety setting (§7).

prime_safe / adaptation / ses_subgame coincide on Kuhn (all target the same ε -adjusted / blueprint floor, and Kuhn’s SES subgame is the whole game). They sit at worst-case $-0.063 = \mathbf{v^*} - 0.008$, flagged “unsafe” *only* because the flag compares to $\mathbf{v^*}$ while they legitimately target a lower floor by design (§8).

A measurement-resolution note. In the **smoke** run (30 000 CFR iters) even the Nash policy is flagged “unsafe” (worst-case -0.0568 , i.e. 0.0013 below the exact $\mathbf{v^*}$, just over the 1e-3 tolerance) — a pure CFR-approximation artifact. At **scale** (200 000 iters) Nash’s shortfall drops to 3e-4 and it is correctly flagged safe. Lesson for the harness: the safe-flag tolerance must exceed the baseline’s own approximation error, or the baseline trips its own test.

3.3 7. The efficient frontier — and the bang-bang surprise

Sweeping the canonical RNR parameter p against TightPassive, alongside the naive blend and the three LP operating points (from **results/pareto_kuhn.json**):

	canonical RNR —	canonical —	naive blend —	naive —
p	EV	exploitability	EV	exploitability
0.0	−0.044	~0.000	−0.047	0.000
0.3	−0.044	~0.000	+0.017	0.133
0.6	−0.044	~0.000	+0.081	0.266
0.7	+0.167	0.444	+0.103	0.311
1.0	+0.167	0.444	+0.167	0.444

Operating points: best equilibrium **ganzfried** (EV -0.044 , exploitability 0.0005); **prime_safe / adaptation** (EV -0.040 , exploitability 0.0079); measured $\varepsilon = 0.0074$.

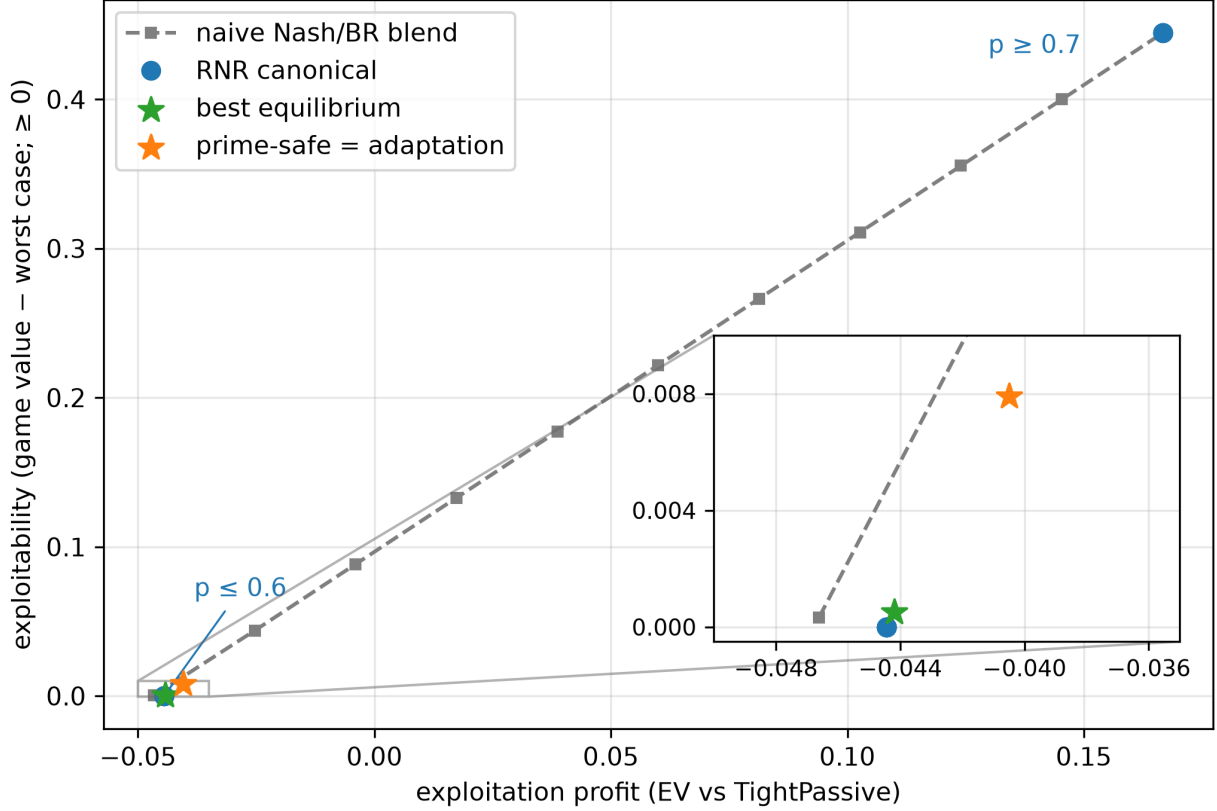


Figure 4: Exploitation-safety frontier (Kuhn). The canonical RNR “curve” is an interpolation between just two achieved clusters (safe corner and full-BR corner); the naive blend traces the smooth line; the LP operating points (stars) sit at the efficient safe corner.

3.3.1 Prediction ↔ reality: canonical RNR is bang-bang, not a smooth frontier

- **Prediction (Phase 4):** “the RNR sweep is monotone, and canonical RNR dominates the naive blend everywhere.”
- **What actually happened:** canonical RNR is a **step function**. For $p \in [0, 0.6]$ it returns the *same* safe strategy (EV -0.044 , exploitability ≈ 0); at $p \approx 0.7$ it jumps straight to the *full* best response (EV $+0.167$, exploitability 0.444). Between the sampled values (steps of 0.1) there are **no intermediate points**.
- **Why (verified reasoning).** The canonical RNR objective $\max_{\mathbf{x}} [p \cdot \text{EV}(\text{model}) + (1-p) \cdot \min_{\{\sigma'\}} \text{EV}(\mathbf{x}, \sigma')]$ is *linear* in the hero realization plan over a *polytope*, so its optimum is a **vertex** that switches only when p crosses a critical ratio. Kuhn’s strategy polytope is tiny (few vertices), so the transition is a single jump. Johanson’s smooth RNR frontier is a large-game / data-biased phenomenon (many vertices, or a per-info-set p); it does not appear in a game this small. The **naive blend** *does* trace a smooth line — precisely because it interpolates two fixed strategies — but it is dominated at the safe corner (at

exploitability ≈ 0 the LP methods reach EV -0.044 vs the blend’s -0.047) and buys its interior profit at a steep exploitability cost.

- **What is bang-bang is the map from p to a strategy, not the frontier.** Mixing the two RNR strategies (Johanson et al. define this mixture) reaches every point on the chord between the corners, because EV is linear in the mixture and the worst case is concave. With the two vertices found, the switch happens at $p \approx 0.68$; any further vertex would be optimal only for p between 0.6 and 0.7, which the sweep did not sample. Against TightPassive the chord improves on the naive blend by only ≈ 0.002 , so in this game choosing *where* to deviate buys almost nothing over uniform scaling (Johanson et al. report strongly concave frontiers in a large abstracted hold’em game). What Kuhn does show is that p is not a dial: the switch threshold moves with how exploitable the opponent is (§6: `rn timer_0.5` is already at full BR vs `AlwaysPass`). This is why best equilibrium — which constrains the *value* rather than a p — is the better primitive.

3.4 8. Prime-safe / adaptation and the measured ϵ -budget

Prime-safe and adaptation lower the safety floor from v^* to $v^* - \epsilon$, where ϵ is the baseline’s own exploitability, measured from an early-stopped CFR run (never fabricated). The run measured $\epsilon = 0.0074$, and the prime-safe/adaptation worst-case came out at $-0.063 = v^* - 0.0079$ across every opponent — matching the ϵ -adjusted floor to rounding. Spending that budget, they earn slightly more than best equilibrium: **+0.266 vs +0.222** against `AlwaysPass`, $+0.151$ vs $+0.131$ against `LooseAggressive` (§6). They are flagged “unsafe” in the table only because the flag references v^* ; against their *own* ϵ -floor they are safe by construction. Prime-safe and adaptation coincide by construction: ϵ is defined as v^* minus the baseline’s worst case, so $v^* - \epsilon$ is that worst case, and the run uses the same early-stopped CFR strategy as the prime-safe baseline and the adaptation blueprint. Jeary & Turrini’s floor and Ge et al.’s coincide whenever the two are the same strategy; they differ in setting (whole-game exploitation versus subgame re-solving), not in the floor.

Conclusion. The prime-safe mechanism works exactly as designed: an imperfect (ϵ -exploitable) baseline is honestly *measured*, and the safety floor is relaxed by precisely that amount — converting Ganzfried and Sandholm’s “needs an exact Nash” guarantee into one usable with the approximate baselines any real system has.

3.5 9. The teaching attack — why realized profit is the wrong lens

What we test. A deceptive opponent plays the weak **TightPassive** bait for the first 10 000 hands, then switches to a strong **Nash** “reveal” for 10 000 more; a Chapter 7 continuous model feeds each solver every 500 hands (5 seeds). Does the safety floor protect the safe methods when the opponent turns on them? From `results/kuhn_scale.json`.

method	mean/hand (all)	mean/hand (after switch)	safety violations / seed
full_br	+0.051	−0.061	40, 40, 40, 40, 40
best eq. (ganzfried)	−0.048	−0.055	0, 0, 0, 0, 0
adaptation	−0.046	−0.055	40, 40, 40, 40, 40
nash	−0.051	−0.061	0, 0, 0, 0, 0

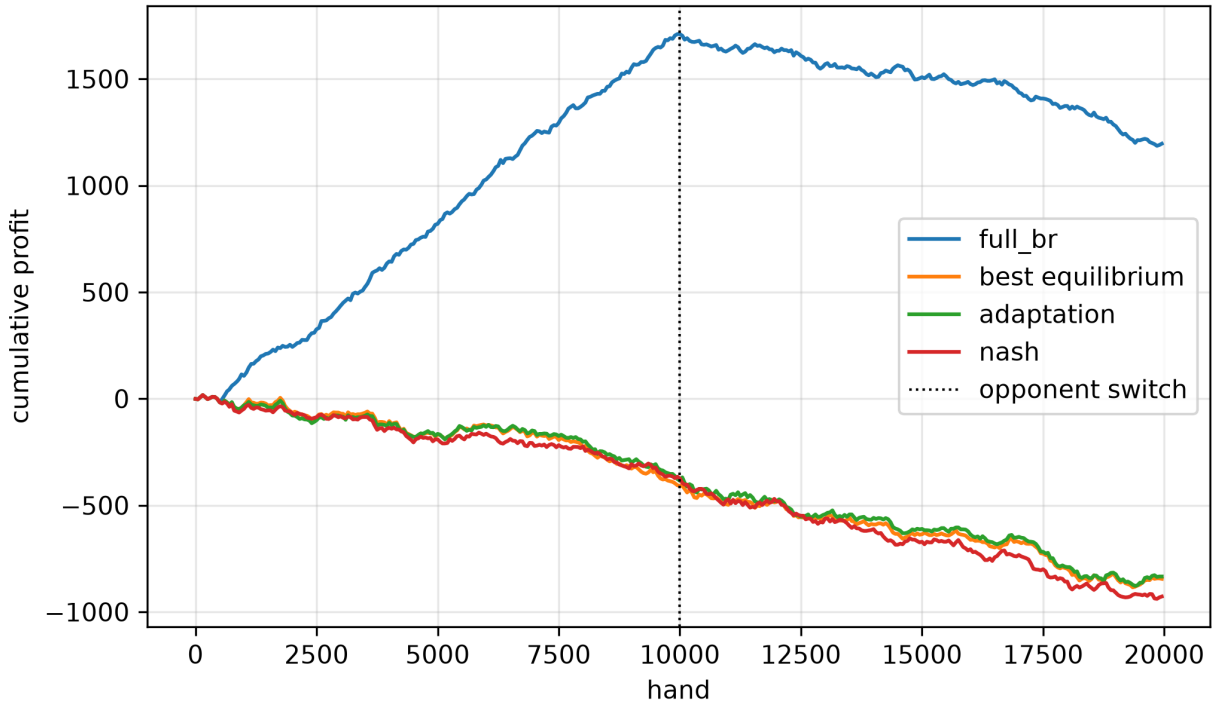


Figure 5: Teaching attack (Kuhn), cumulative profit (the run with seed 0). `full_br` (blue) climbs on the bait to $\approx +1700$, then only drifts down after the switch — it ends far ahead because a Nash “revealer” claws back only $\approx$ the game value per hand. The safe methods refuse the bait and pay the P0 tax throughout.

3.5.1 Prediction $\leftrightarrow$ reality: a Nash revealer is too gentle to punish full_br

- **Prediction (Phase 4):** “*safe methods’ post-switch mean stays near baseline; full_br’s is clearly worse; safety violations = 0 for safe methods, > 0 for full_br.*”
- **What actually happened.** The *safety-violation* half held **cleanly**: full_br violated the Nash floor at **40/40** refits, best equilibrium (**ganzfried**) at **0/40**. But full_br’s post-switch rate (-0.061) is only *marginally* worse than the game value (-0.056), and it ends **hugely net-positive** overall ($+0.051$ /hand) — because the “reveal” opponent is *Nash*, which only wins back $\approx$ the game value per hand, so the bait-phase windfall is never clawed back within 10 000 hands.
- **Why (verified reasoning).** A strategy’s worst-case is realized by an opponent that *best-responds to it* (that is what drives full_br to -0.5 in §6). A stationary Nash reveal is **not** that adversary, so the online realized-profit experiment never triggers the risk that the exact worst-case column exposes. Measuring the teaching attack by profit-vs-Nash therefore *understates* the danger.
- **Corrected takeaway.** The honest separating signal here is the **exact worst-case / safety-violation count** (full_br 40, best equilibrium 0), not realized profit. To make the teaching attack punish full_br *in profit*, the reveal must be an **adaptive counter-exploiter** (a best response to the exploiter’s stale model), not a fixed Nash. That is the concrete refinement for the next run (§13).
- **A design nuance, not a bug.** adaptation shows 40 violations while nash shows 0 (at scale): adaptation deliberately targets a floor *below* **v*** (§8), so its played strategy trips the **v***-referenced counter every refit — by design. Nash’s 0 (at scale) confirms the smoke-run’s 10 “violations” were the CFR-resolution artifact of §6.

3.6 10. Leduc — the headline finding: within a 40-iteration cap the global solvers do not converge; SES (cap 400) does

The Leduc run is the human-extended **bounded_scale** config: a **40-iteration cap** on the constraint-generation loop, `leduc_tol = 0.01`, and the SES subgame set to the King-flop (`leduc_flop_rank(King)` — exploit only after a King appears on the board). Each cell records whether the solve **converged** or hit the cap (**capped**). From `results/leduc_bounded_scale.json`; Nash floor **v*** = -0.086 .

Opponent metric		nash	full_br	ses_subgame	best eq.	prime_safe	adaptation
Rock	EV	+0.201	+0.937	+0.247	+0.624	+0.635	+0.635
	worst-	-0.089	-1.633	-0.130	-0.838	-0.744	-0.744
	case						

Opponent metric	nash	full_br	ses_subgame	best eq.	prime_safe	adaptation
Maniac	converged?	✓	✓	✓ (194 it)	✗ capped	✗ capped
	EV	+0.438	+2.177	+0.682	+1.806	+1.842
	worst-case	−0.089	−1.100	−0.130	−0.638	−0.657
LoosePassive	converged?	✓	✓	✓ (211 it)	✗ capped	✗ capped
	EV	+0.449	+1.405	+0.559	+1.306	+1.309
	worst-case	−0.089	−4.200	−0.133	−1.239	−1.334
CallingStallion	converged?	✓	✓	✗ (400 it)	✗ capped	✗ capped
	EV	+0.559	+1.464	+0.663	+1.342	+1.363
	worst-case	−0.089	−1.000	−0.130	−0.915	−0.686
Nash (control)	converged?	✓	✓	✓ (350 it)	✗ capped	✗ capped
	EV	−0.086	−0.083	−0.091	−0.083	−0.083
	worst-case	−0.089	−0.842	−0.129	−0.376	−0.325

3.6.1 Prediction ↔ reality: global safety does not scale to Leduc as written

- **Prediction (Phase 4):** “*Ganzfried: worst-case $\geq v^*$ within tolerance (safe).*” And the implementation README’s #1 likely-to-break item: “*constraint-generation convergence + LP conditioning.*”
- **What actually happened.** On Leduc the **global** solvers (best equilibrium ganzfried, prime_safe, adaptation, rnr_0.5) **all hit the 40-iteration cap without converging**, leaving worst-case values of **−0.64 to −1.33** (safety violations of 0.55–1.25) — grossly unsafe. The Kuhn-validated “best equilibrium is safe” result **did not carry over** to Leduc within a 40-iteration budget.
- **Why (verified reasoning).** The cutting-plane loop adds one adversary best-response cut per iteration; on Leduc’s larger tree the set of relevant pure best responses is large, so 40 cuts are nowhere near enough to pin the true worst case, and the master keeps returning optimistic-but-unsafe strategies. This is precisely the failure mode flagged before the run — now confirmed with numbers.
- **The positive result.** The **subgame method (SES)** **converged** on 3 of 4 exploitable opponents (194 / 211 / 350 iterations), because it re-solves only the King-flop subgame with the rest of the tree pinned to the blueprint — a far smaller LP with a far smaller adversary

set. It extracts real value (**+0.25 to +0.68** vs the weak types, beating Nash) at worst-case ≈ -0.13 , an **order of magnitude closer to safe** than the global methods (-0.13 vs $-0.64\dots-1.33$). The comparison is not at equal budgets, though: the global solvers were stopped after 40 iterations (about 2.5 s per cell), while SES had a cap of 400 and used 194–400 iterations (30–79 s per cell). The result shows the global loop is slow and unsafe *within this budget*, not that it cannot reach safety with a larger one. Chapter 14 settles the scaling question: its one-shot dual LP, which replaces the loop, solves the full-Leduc problem in 0.02–0.04 s, so the wall was the loop’s, not the global problem’s.

- **Honest caveat on SES.** Its residual exploitability (≈ 0.043) still exceeds the 0.01 Leduc tolerance, so it too is *flagged unsafe*, and on **LoosePassive** it ran 400 iterations without converging. That violation is measured against **v***, though. SES’s own floor is the blueprint’s worst case — here the early-stopped CFR baseline (-0.120 in the run notes), not the Nash strategy (-0.089) — and the three converged cells sit 0.010 below it, exactly at the 0.01 tolerance. So SES held its gadget floor to within the tolerance, and the residual is mostly the blueprint’s own exploitability; a rerun with a tighter tolerance would show whether it is *provably* safe.
- **What this means.** This is the **global-vs-local safety** distinction — introduced in the intuition and reading phases as theory — appearing *empirically* at a scale as small as Leduc. It is the concrete, measured motivation for real-time subgame methods (SES / OX-Search) and for replacing the cutting-plane loop with an exact one-shot dual LP. A testbed this small was large enough to reveal the scaling wall.

3.6.2 Solver hardening the runs forced

The Leduc results carry fields the Phase-4 code did not emit, so the solver was hardened during execution — matching the anticipated debugging: a small feasibility slack on the cuts (**_FEAS_SLACK** = **5e-4**, so an approximate-Nash floor that slightly overshoots the true max-min does not make the master infeasible), explicit infeasibility handling (keep the last feasible strategy), per-cell **converged** / **capped** / **iterations** tracking, and the **bounded_scale** Leduc driver with a King-flop SES predicate. These are exactly the mitigations the implementation README’s “likely to break” list pointed at.

3.7 11. Are the results trustworthy?

Three independent checks support the numbers, with two caveats.

- **The exact anchors reproduce known theory.** The measured Kuhn game value is **−0.0556** ($\approx -1/18$, the known first-player disadvantage ✓); the Leduc value is -0.0862 . Against the Nash opponent every method earns $\approx$ the game value and no more (§6, §10) —

as it must, since an equilibrium cannot be exploited; where a number sits a hair above its ceiling it is finite-sample noise, not a real breach.

- **Internal consistency of the LP engine.** Every solver that should reach the full best-response value does: `full_br` and post-transition `rn` reach exactly +0.167 (Kuhn TightPassive) and the large Leduc BR values, which is only possible if the sequence-form payoff vector $c \cdot x$ matches the exact best response — the engine’s central correctness condition.
- **Convergence is now observable.** The `converged` / `capped` flags (Leduc) make the one place the method can silently fail — constraint-generation non-convergence — explicit in the output rather than hidden.

Caveats. (1) No `validate.py` PASS/FAIL log and no OpenSpiel cross-check artifact were saved, so those two harness outputs are unverified from artifacts here (re-run to close). (2) Leduc was run only in the iteration-capped `bounded_scale` config, so the global solvers’ Leduc numbers reflect a *budget*, not a converged solve — they show the method is slow/unsafe **within 40 iterations**, not that it can never reach safety given far more.

3.8 12. Limitations

Ranked by how much they qualify the conclusions:

1. **Global safe-exploitation did not converge on Leduc (§10).** Under a 40-iteration cap the cutting-plane loop leaves best equilibrium/prime-safe/adaptation grossly unsafe. This is the clearest gap and the main research hook — resolve with an exact dual-LP or a much larger budget, and confirm whether it is “slow” or “structurally stuck.” (Chapter 14’s one-shot dual LP resolves it: full Leduc in 0.02–0.04 s.)
2. **SES residual exploitability (§10).** SES converges; its ≈ 0.04 violation is measured against v^* , while against its own floor (the blueprint’s worst case) the converged cells sit exactly at the 0.01 tolerance. Whether it is *provably* safe needs a rerun with a tighter tolerance.
3. **The teaching attack under-punishes (§9).** A stationary Nash reveal does not trigger the worst-case, so realized profit fails to separate safe from unsafe — only the violation count does. An adaptive punisher is needed to show a safe method *out-earning* `full_br` under deception.
4. **Two small games; perfect models in the exact table.** The §6/§10 tables solve against a *perfect* model of each type (the offline case); the online, learned-model case is only the teaching attack. Kuhn/Leduc are exact-solvable *because* they are tiny; the convergence wall that appears on Leduc will only worsen at scale — this is a controlled proof-of-concept, not a scaling claim.
5. **Unverified cross-checks (§11).** `validate.py` and the OpenSpiel comparison were not captured as artifacts.

3.9 13. Conclusions and research directions

Conclusions. Safe exploitation is one idea — *maximize value against the model subject to a safety floor* — and on a fully-solvable game it works exactly as the theory says: **best equilibrium (the Ganzfried–Sandholm baseline) is safe against every opponent and earns more than Nash on every exploitable one** (+0.222 vs +0.146 against AlwaysPass), if only by 0.002–0.076 per hand, where naive best response earns more but is ruinously exploitable (worst-case -0.5). Prime-safe/adaptation extend this to imperfect baselines by spending a *measured* ε -budget below v^* . But the chapter’s most valuable result is negative and empirical: on a game as small as **Leduc**, the *global* safe-exploitation solve did **not** converge within a 40-iteration cap, while the *local* subgame method (SES, cap 400) did — the global-vs-local safety gap, measured at unequal budgets. Two Phase-4 predictions (smooth RNR frontier; best equilibrium safe on Leduc) were contradicted and reconciled; the reconciliations are more instructive than the predictions would have been.

Research directions (each motivated by a measured effect above):

- **The stronger two-player baseline (§1).** Implement Ganzfried & Sandholm’s gift-risking RWYWE — the existing LP with a per-hand floor $v^* - k_t$ that moves with the banked gifts — as the baseline for Contribution #2, and rerun the Kuhn/Leduc comparison, with the Leduc global and subgame solvers at equal budgets. Chapter 15 implements and measures RWYWE (results in §1); Chapter 14’s exact dual LP supersedes the equal-budget rerun.
- **Scalable safety — the central next chapter (§10).** Replace the cutting-plane loop with the **exact one-shot dual LP** for the worst-case constraint, *or* commit to **local / subgame** safety (SES / OX-Search) as the scalable path. The Leduc non-convergence is direct evidence for the latter. Chapter 14 took the first path: its one-shot dual LP solves full Leduc in 0.02–0.04 s.
- **Resolve the SES gadget’s residual exploitability (§10).** The 0.04 gap is measured against v^* ; against its own blueprint floor SES sits at the 0.01 tolerance. A rerun with a tighter tolerance decides whether the subgame method is provably or only approximately safe.
- **A punishing teaching attack (§9).** Replace the Nash reveal with an adaptive counter-exploiter so realized profit corroborates the worst-case view, and a safe method can be shown to *out-earn* full_br under deception.
- **N-player safety — thesis Contribution #2.** Every guarantee here rests on the two-player zero-sum fact that a Nash strategy secures v^* against any opponent; for $N > 2$ there is no such v^* anchor. N-player safety notions exist but are either very conservative (maxmin against a coordinated coalition, Celli & Gatti 2018) or not securable in general (equal share, a fair-share

target that cannot be secured against opponents with different fixed strategies, Ge et al. 2025). The open thesis problem this chapter makes precise is exploitation with a loss bound relative to a baseline, tested also against colluding opponents.

- **Close the validation loop (§11).** Run and archive `validate.py` and the OpenSpiel cross-check; re-run the Leduc global solvers with a large budget / exact dual to distinguish “slow” from “structurally unsafe.”

3.10 14. Reproduction

Part I — exploration (from `implementation/step08/exploration/`, `.venv` active; each prints tables to stdout and writes figures under `exploration/figures/`):

```
python pareto_curve.py           # Exp 1 - naive Nash/BR frontier
python rnr_playground.py        # Exp 2 - naive RNR p-sweep
python exploitation_safety_playground.py # Nash vs BR vs blend, all metrics
python naive_exploit_danger.py   # why full BR is dangerous
python subgame_peek.py          # blueprint vs local exploit (Leduc)
```

Part II — full evaluation (from `implementation/step08/implementation/`, `.venv` active; needs `numpy`, `scipy`, `matplotlib`):

```
python tournament.py --config smoke # quick Kuhn pass (~1.4 s)
python tournament.py --config scale # Kuhn full run (~10 s): table + teaching attack
python pareto.py --config scale     # the exploitation-safety frontier
python validate.py                 # PASS/FAIL vs the raw-step targets
python compare_openspiel.py        # OpenSpiel NashConv cross-validation (needs open_spiel)
# Leduc used a human-added bounded_scale driver (40-iter cap, King-flop SES); see results/leduc
```

Results write to `results/*.json` and plots to `plots/*.png`. Figures reproduced in this report live in `deliverables/reports/step08/figures/` (exploration figures under their original names; evaluation figures prefixed `impl_`). The full verified consolidation, with the prediction→reality ledger, is at `implementation/step08/consolidation/report.md`.